Cabin Lifecycle DFDs

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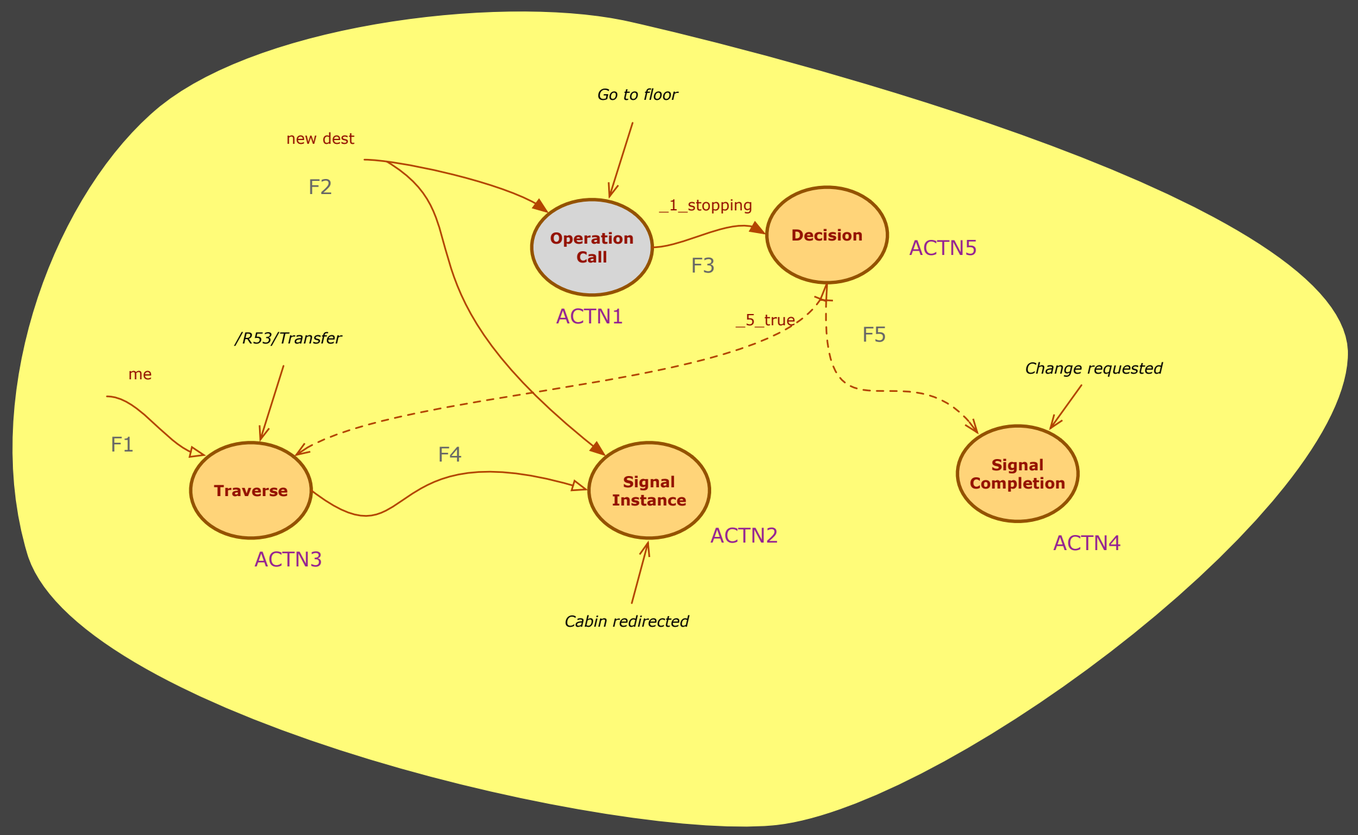
Changing destination

```
~.Go to floor( Dest floor: ^new dest )? {
   Cabin redirected( Updated dest: ^new dest ) -> /R53/Transfer
   Change requested -> me
}
```



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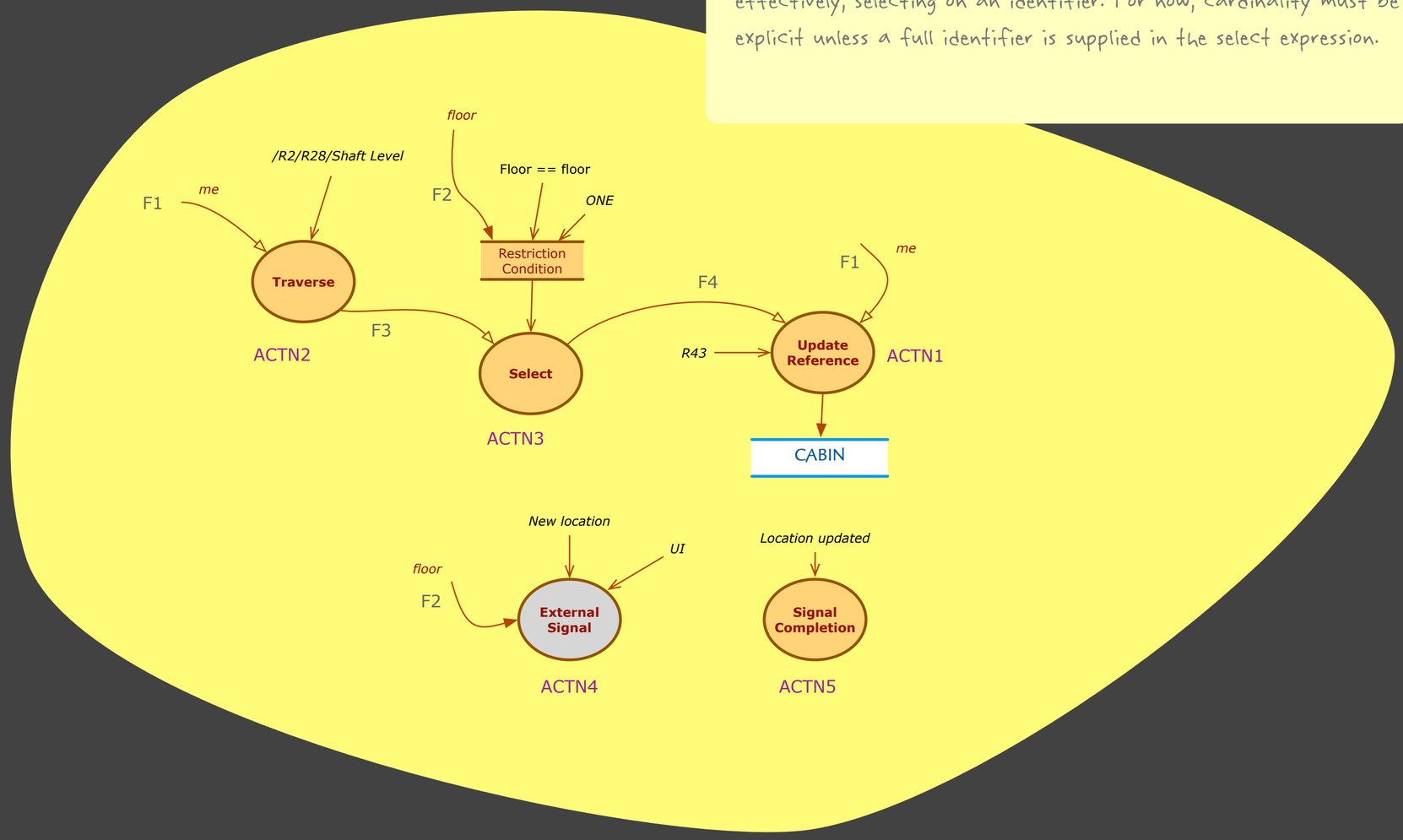
Update location

&R43 /R2/R28/Shaft Level(1, ∧floor)
New location(∧floor) -> ~UI
Location updated -> me

We require the explicit 1 cardinality in the Shaft Level selection even though it could be derived. But the populate isn't smart enough to do that yet. It should be seeing that since we traverse from a single instance of Cabin to a single instance of Shaft to many instances of Shaft Level and that the identifier of Shaft Level is Shaft + Floor with Shaft implicitly provided by the navigation with a single value of Floor specified that we are, effectively, selecting on an identifier. For now, cardinality must be explicit unless a full identifier is supplied in the select expression.



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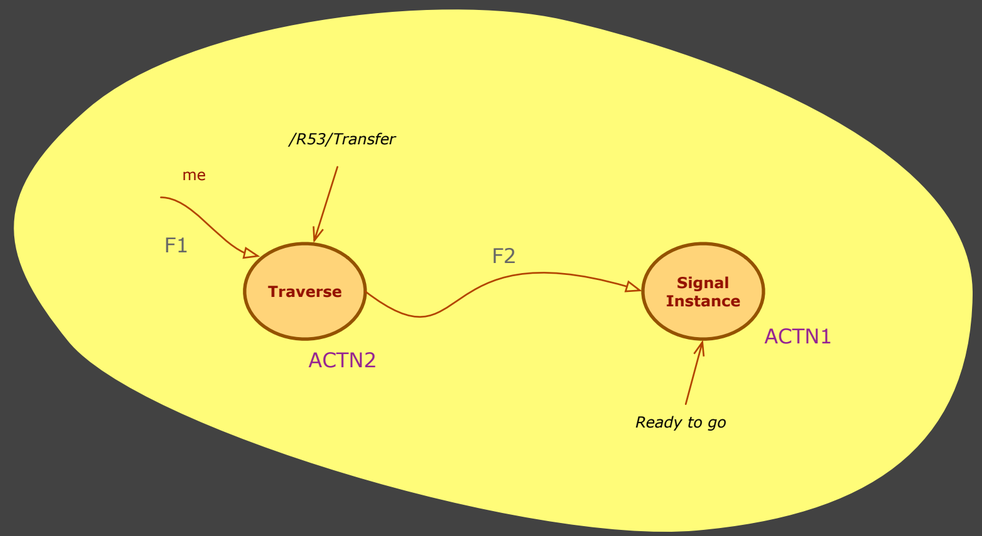
A30

READY TO GO

Ready to go -> /R53/Transfer



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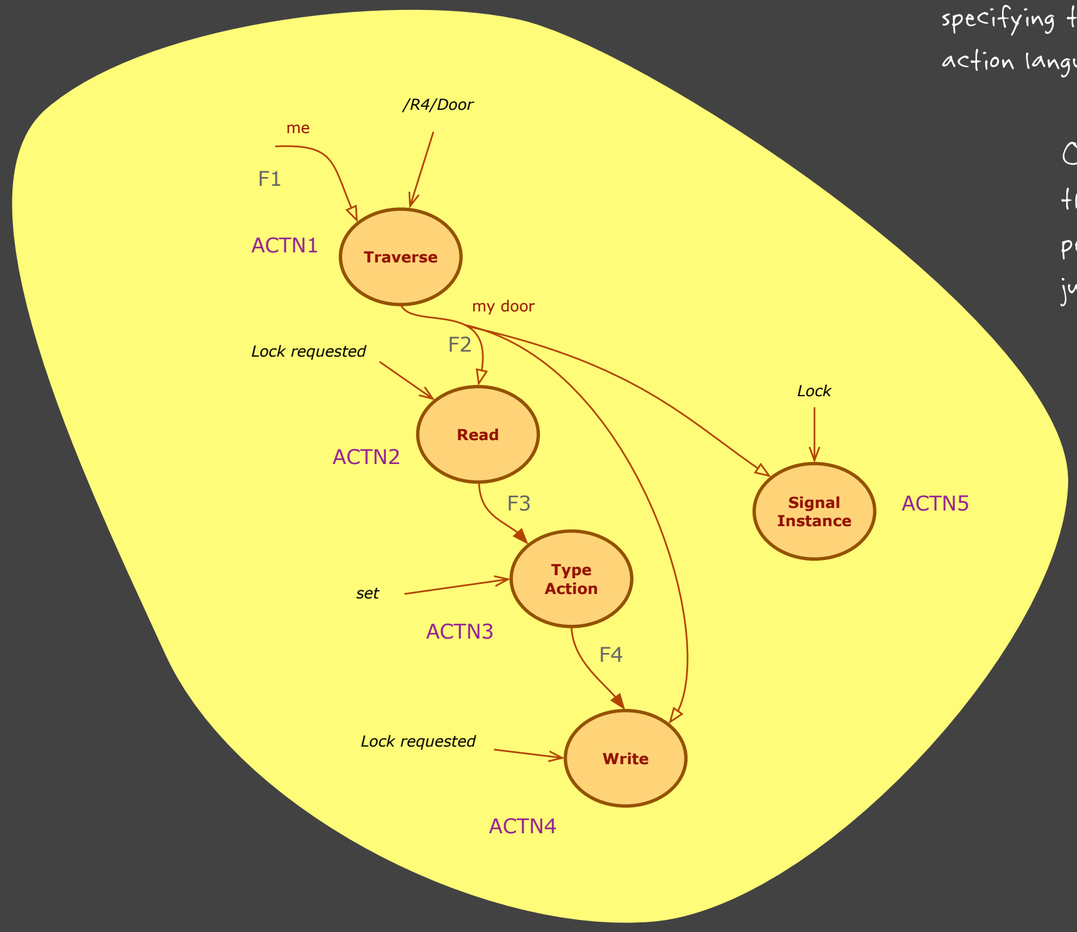
SECURING DOORS

```
my door .= /R4/Door
my door.Lock requested.set
Lock -> my door
```



By assigning the traversal result to a labeled flow, we avoid specifying two identical traverse actions as we do with the alternate action language (see next page).

Of course we could post process the action recognizing and replacing the duplicate actions. And in the future, we should definitely perform flow graph optimizations prior to execution. For now, let's just simplify the action language we write.

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Requesting transport

Go to floor(dest floor: /R53/Transfer.Destination floor) -> ~TRANS
Transport in progress -> me

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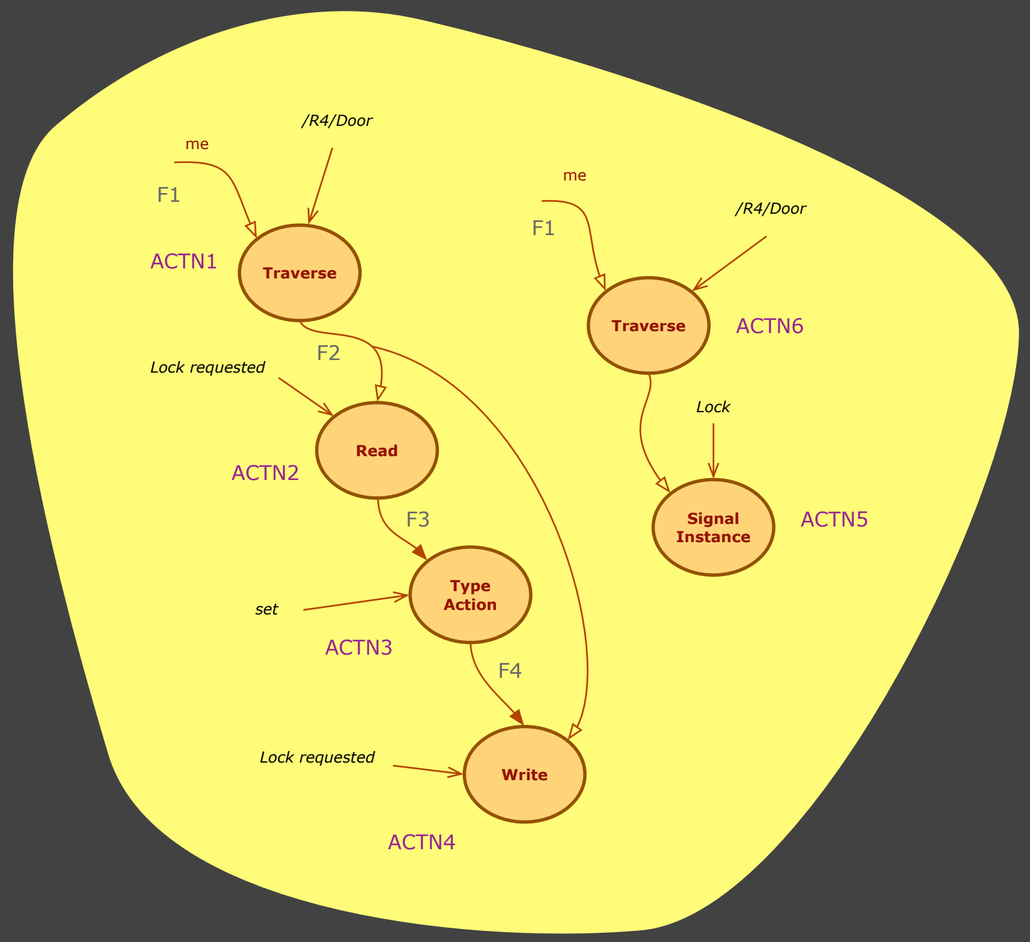
SECURING DOORS

/R4/Door.Lock requested.set
Lock -> /R4/Door



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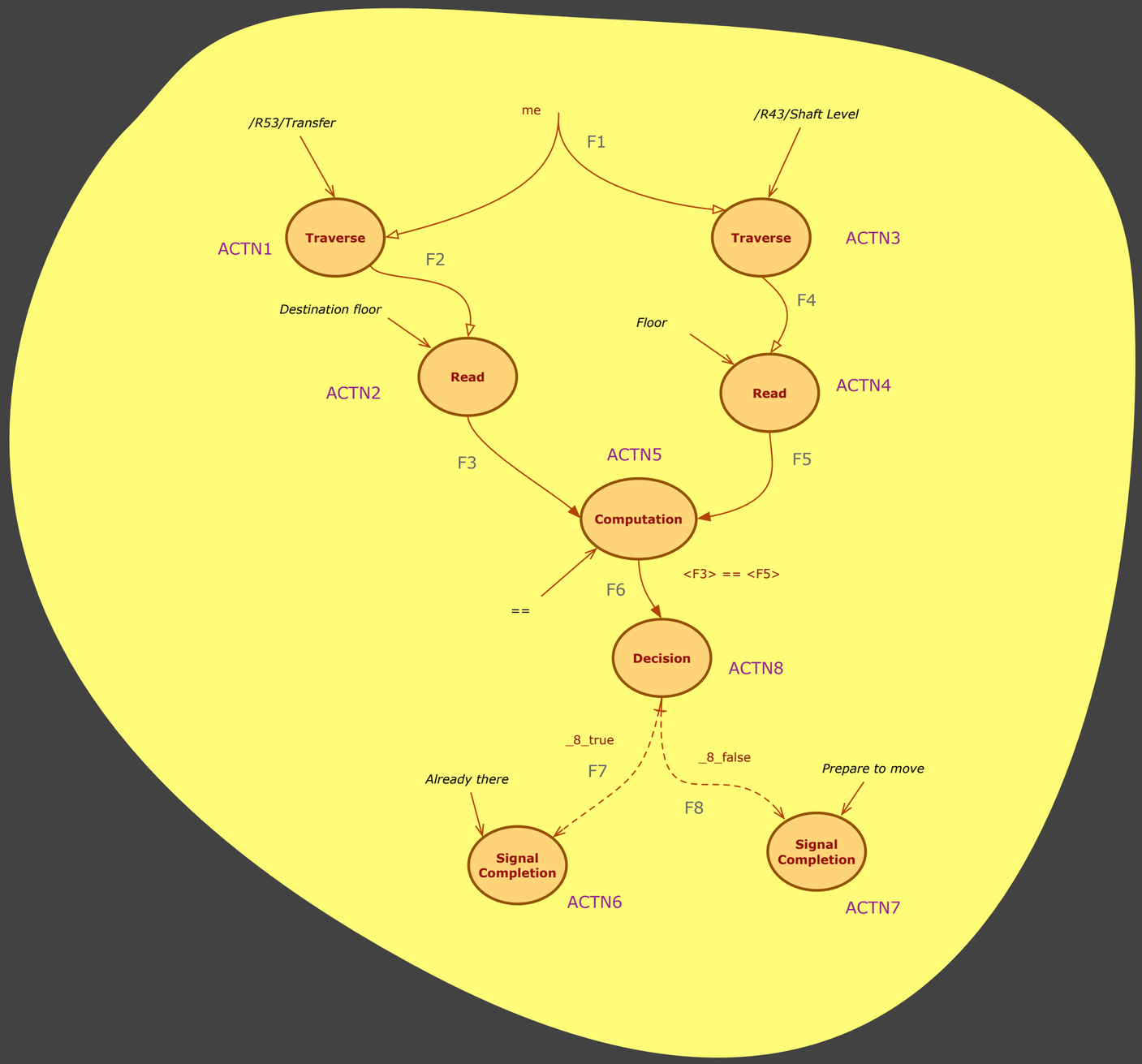
Are we already there?

/R53/Transfer.Destination floor == /R43/Shaft Level.Floor ?
Already there -> : Prepare to move -> me



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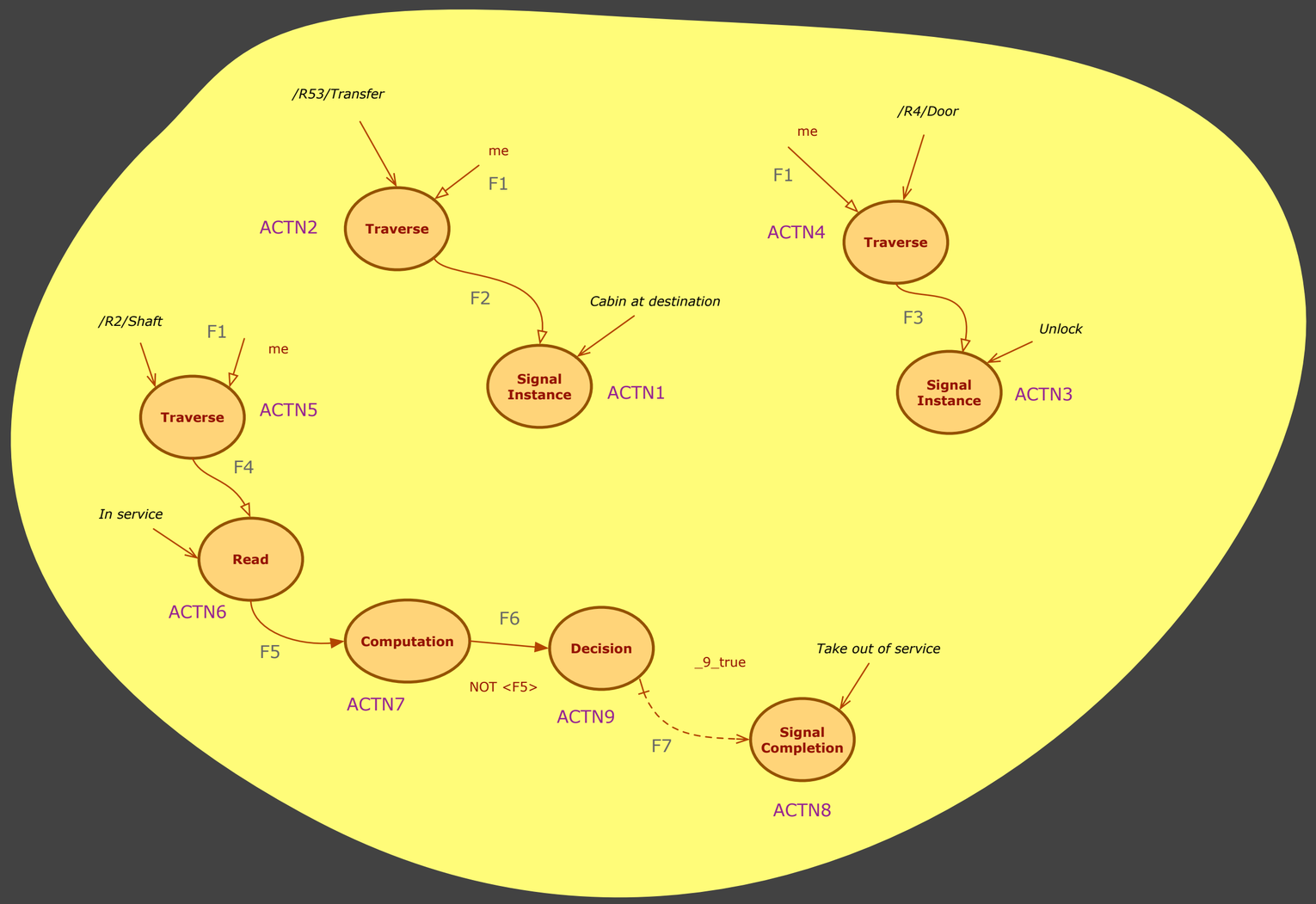
PICKUP DROPOFF

Cabin at destination -> /R53/Transfer
Unlock -> /R4/Door
! /R2/Shaft.In service ? Take out of service -> me



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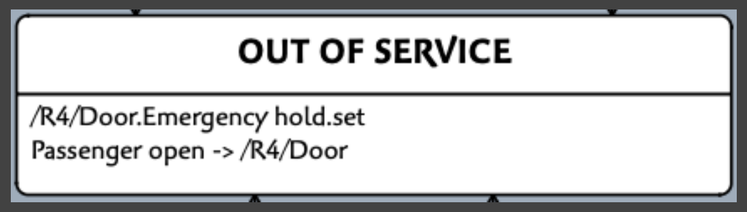

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